

MEMORY STOCKS

Industry Trends and Opportunity

Memory has always been a foundational layer of the AI infrastructure stack, but for much of the early buildout it was an afterthought, overshadowed by insatiable demand for CPUs and GPUs. That shifted decisively as the industry pivoted toward agentic architectures and inference-time scaling.¹ Unlike training workloads, where data can be batched across long time horizons, agentic systems require persistent context, real-time retrieval, and sustained lowlatency memory access across multi-step reasoning chains. This is where dynamic randomaccess memory (DRAM) and high bandwidth memory (HBM) become critical, each serving a distinct role. DRAM acts as the system-level working memory, while HBM, an evolution of DRAM stacked vertically and integrated directly onto the processor package, sits millimeters from the GPU itself, delivering the extreme bandwidth needed to feed AI accelerators midcomputation. As models grew deeper and inference demands more stringent, both tiers saw explosive demand growth, with HBM in particular transitioning from an optional premium to hard architectural requirement. This structural re-rating of memory's role across the AI stack is what drove much of the outsized returns across the semiconductor industry over the past 12 months. Additionally, such demand has been reflected in memory pricing, which has experienced unprecedented rates of growth over the past year. (See Exhibit 1).

Memory operates as a commodity market, with prices set by the balance of supply and demand rather than by any individual producer's pricing power, and the three major suppliers sell functionally the same products. However, it also has extraordinary barriers to entry that have produced a stable oligopoly. Building a leading-edge memory fab requires more than \$20 billion in upfront capital, years of construction, and process know-how refined over decades, a combination that has kept any credible new entrant from emerging in more than 20 years. The result is a rational oligopoly: Samsung, SK Hynix ("SK"), and Micron (MU) have every incentive to discipline supply rather than chase share. For the DRAM market, SK holds a 34 percent share, Samsung 33 percent, and MU 26 percent. For HBM, MU holds a ~21 percent share, roughly tied with Samsung, while SK dominates at 58 percent.

Within this, there are also inherent qualification risks for each company. If they do not qualify for the necessary CPU/GPU customers—such as Nvidia (NVDA), Advanced Micro Devices (AMD), and others—they are shut out of production. Qualification is the formal validation process by which a memory supplier proves its product meets a CPU/GPU customer's electrical, thermal, and reliability specifications on their reference platform. It is conducted per product and per platform—qualifying on NVDA's Blackwell (HBM3E) does not carry over to their Rubin (HBM4) platform, and AMD's MI300 qualification does not transfer to MI350, for

¹ Agentic architectures are AI systems where models autonomously plan, use tools, and execute multi-step tasks rather than producing oneshot responses; inference-time scaling improves output quality by allocating more compute at the moment of answering (e.g., extended reasoning, multiple solution attempts) rather than only during training.

example. The process moves through sample delivery, characterization, reliability stress testing, and production qualification at volume, typically running 9 to 18 months end-to-end. Missing qualification on a flagship platform excludes a supplier from that entire generation's lifecycle, while qualifying late usually caps allocation at a 10 to 25 percent minority share with weaker pricing power. Recent cycles have shown both outcomes; suppliers narrowly clearing qualification to secure allocation, and others failing outright and forfeiting billions in addressable revenue.

These dynamics have historically produced a brutal cyclicalality on both a company and sector level, whereby demand spikes drive producers to aggressively expand production capacity and supply, that supply oversaturates the market, prices collapse, and margins evaporate. The memory industry has lived and died by this cycle for decades, and the central debate today is whether AI represents a genuine structural break from that pattern—a permanent change in demand driven by large language models, agentic systems, and accelerated compute infrastructure—or simply another cyclical upturn. Bulls argue the sheer scale and sustained growth of AI workloads creates a demand floor the industry has never seen before, while bears point to the same overcapacity risk that has humbled memory investors in every prior cycle. The answer will define the memory trade for the next several years.

Beyond cyclicalality, the deeper risk within the memory trade is the inherent unpredictability of innovation amid the rapidly evolving AI landscape. The architectural requirements of AI systems have shifted dramatically even within short windows—memory was largely an afterthought just one year ago before agentic and inference-time scaling rewrote the demand calculus entirely. That same velocity of change cuts both ways. Google's TurboQuant research, which proposes a method to reduce LLM memory consumption by up to 6x, is a pointed example of innovation risk within the sector. Because the same pace of AI development that created the memory supercycle is capable of disrupting it, investors are not just betting on sustained AI growth, but on the specific memory-intensive architectural stack that underpins it.

MICRON TECHNOLOGY (MU)

Company Overview

MU is one of the largest semiconductor companies in the world, designing and manufacturing the memory technology that powers everything from smartphones and laptops to the AI infrastructure driving the current wave of technological development. Revenue breaks down approximately 57 percent DRAM, 22 percent HBM, and 21 percent flash (NAND) memory (local storage that holds the bulk of a device's data and retains it while powered off). HBM is MU's fastest growing and highest margin product category by a wide margin. Although the technology has existed since 2013, MU was a late entrant and held negligible share throughout earlier generations; the segment was effectively non-existent three years ago and only became material once MU qualified its latest-generation HBM with NVDA. Now, HBM is the primary

growth driver of the entire company. Within the memory industry, MU is the third largest player in a three-way oligopoly, with SK Hynix and Samsung controlling the majority of global DRAM and HBM supply.

While DRAM has been the focal point of the memory upcycle, NAND deserves mention as well. NAND prices have actually risen faster than DRAM in recent quarters, driven by similar AI-related demand for enterprise solid state drives (SSDs) in data centers, supply discipline from the major producers, and a multi-year underinvestment in new capacity following the 2022–2023 downturn. Like DRAM, NAND is a consolidated oligopoly; Samsung, SK Hynix (which acquired Intel's NAND business), MU, Kioxia, and Sandisk (SNDK) account for nearly all global supply.

Outlook & Competitive Advantages

MU is well positioned in a rapidly expanding market with demand that currently outstrips all available supply. As noted in Exhibit 2, the four largest hyperscalers in the world, Amazon, Microsoft, Google, and Meta, are collectively projected to spend about \$3.8 trillion in capex (about 60 percent of that specifically on hardware/semiconductors²) from 2025 to 2030, the single largest coordinated capital deployment in the history of the technology industry. With only three companies capable of supplying the HBM and DRAM that powers this infrastructure, MU sits in an extraordinarily advantaged position, even as the smallest player. In today's unpredictable geopolitical environment, MU's status as the sole US-based manufacturer of DRAM and HBM is an increasingly important strategic and regulatory differentiator, reflected in CHIPS Act tax credits the company has accumulated. SK, Samsung, and SNDK all lack US fabs, instead operating out of China, Korea, and Japan, respectively.

MU's entire 2026 HBM and DRAM supply is sold out under locked price and volume contracts, with the company filling only 50 to 67 percent of customers' HBM demand and a similarly constrained share of server DRAM demand. Allocation is prioritized to strategic customers under long-term agreements, such as NVDA and the top hyperscalers, with smaller buyers pro-rated against forecasted demand or pushed to the spot market at steep premiums.

With no new fabrication capacity coming online until mid-2027 at the earliest, MU's current and future inventory is essentially sold out through 2027.

Valuation Analysis

With a ~\$480 billion market cap, MU currently trades at a trailing P/E of ~19x, against a 2-year average of 30x. Yet against rapidly expanding earnings, MU's forward P/E of ~5x looks far more modest and well below the 2-year average of 10x. Consensus believes MU could achieve remarkable earnings growth of over 600 percent YoY in FY 2026, besting FY 2025's growth rate of ~540 percent. FY 2027 projections currently show earnings growing at a high double digit—if not triple digit—rate once again, assuming present demand dynamics persist

² Alphabet Inc. Q4 2025 Earnings Call, February 4, 2026. CFO commentary. Other sources (CreditSights, Bernstein) est 55-65%

HEADER REMOVED

for the foreseeable future. In some ways, MU's forward-looking P/E may actually be conservative relative to the company's medium-term growth and semiconductor peers, many of which have traded to much loftier valuations despite inferior earnings. Whether that discount is rational or an opportunity is a key question, but if the sector-wide momentum continues MU could be a primary beneficiary with meaningful upside remaining.



SENSITIVITY ANALYSIS REMOVED

While this base case is compelling, it is predicated upon demand durability that is structurally uncertain, with substantially asymmetric downside due to overcapacity risk. MU, SK, and Samsung are all simultaneously expanding fabrication capacity in response to current demand signals, with MU's new greenfield fabs in Idaho and New York not coming online until 2027 at the earliest. If AI infrastructure spending moderates before that capacity arrives—whether from a broader capex pause among hyperscalers, a demand-side efficiency breakthrough such as Google's TurboQuant, or simply the industry's well-documented tendency to overshoot on supply—MU could face a simultaneous compression on both earnings and multiples. Not to mention memory stocks tend to start dropping in price even before cyclical tops, due to smart seasoned investors. In a downcycle, memory P/E multiples historically trough in the 8 to 12x range on peak-cycle earnings that themselves may collapse by 60 to 80 percent. Applying those parameters to MU's current earnings trajectory implies a scenario where shares retrace a substantial portion of their gains well before the recovery cycle begins.

SK HYNIX (000660 KS)

Company Overview

SK Hynix (“SK”) is a Korean-based semiconductor manufacturer and the world's secondlargest memory chipmaker, specializing in DRAM, NAND, and HBM. It is the dominant HBM supplier to AI accelerators, holding roughly 60 percent of the global market. The company's revenue breaks down as follows: DRAM (76 percent, of which HBM represents approximately half), NAND (23 percent), and other (1 percent). As of this writing, SK has a market cap of ~\$540 billion.

SK shares many of the same qualities and risks as MU, since both sell into the same end markets. However, Korean equities often trade at a structural discount to US peers given opaque governance tied to the “chaebol” system, where founding families of companies like SK, Samsung, and Hyundai maintain control through complex cross-shareholdings that have a record of prioritizing family interests over those of minority shareholders. SK also carries additional geopolitical risk, given that its headquarters and primary HBM/DRAM manufacturing complex in Icheon sit roughly 50 miles from the North Korean border. Additionally, the company's DRAM fabrication facility in Wuxi, China currently operates

under a US-validated waiver that allows continued equipment imports despite export controls; that waiver is reviewed annually and vulnerable to revocation or tighter restrictions.

Outlook & Competitive Advantages

SK is Korea-based and currently holds majority share of the DRAM and HBM markets, supported by a deep domestic talent pool and an entrenched memory and semiconductor ecosystem that is difficult to replicate elsewhere. Korea's dominance in memory is not an accident of geography—it is the product of decades of deliberate industrial policy, government-backed R&D consortiums, and a vertically integrated supply chain stretching from raw materials and specialty chemicals to photolithography equipment servicing and advanced packaging. The Korean government has designated semiconductor manufacturing a strategic national industry, and this designation carries tangible consequences, including preferential access to infrastructure, expedited permitting for fab construction, and coordinated talent pipelines through top-tier engineering programs at Seoul National University, KAIST, and Yonsei. SK and Samsung between them employ a significant portion of Korea's most specialized semiconductor engineers, and the density of adjacent equipment suppliers within a tight geographic radius around Icheon and Hwaseong creates compounding productivity advantages that foreign competitors cannot easily replicate by building a fab elsewhere and staffing it from scratch. This ecosystem depth is a meaningful structural moat and compresses development cycles, reduces equipment lead times, and allows SK to iterate on process generations faster than peers operating in more fragmented supply environments. For memory in particular, where process node transitions are measured in months and qualification timelines are unforgiving, that speed advantage translates directly into market share.

Like MU, SK has sold all of its 2026 memory output—HBM, DRAM, and NAND—under signed contracts that lock in volume and price. SK also has more of its 2027 capacity committed than peers. These contracts are not “money in the bank,” however—revenue only books when product ships—and they include standard protections that let either side back out in extraordinary situations (or may default to defined price/volume floors).³ That said, SK has outperformed peers across every HBM generation, from its first iterations in 2018 to the current generation. This is due in part to SK's first-mover advantage and early strategic bet on the technology, having co-invented HBM with AMD in 2013 and investing in production throughout multiple market cycles. SK's proprietary packaging technology, called MR-MUF, also delivers better yields and thermals than competitors' methods. Additionally, the company's entrenched position in NVDA's design process gives SK earlier visibility into next-generation design requirements (each generational win compounds into the next; winning generation “N” gets you earlier access to generation “N+1,” etc.) As a result, SK has consistently reigned as the leading HBM supplier, and this incumbency often softens the penalty for stumbling. For instance, when SK encountered yield and performance challenges on HBM in late 2025 after NVDA raised the spec mid-cycle to demand higher speeds for the Rubin platform, NVDA did not cut them outright. Instead, the company paced the platform schedule around SK to fix the issues and brought Samsung into dual-source roughly 30 percent of the allocation. In contrast,

³ Volume is typically committed within a floor-to-ceiling band (customers must take the floor, suppliers must be able to deliver the ceiling), and pricing is usually fixed on the base volume with a formula-driven structure on incremental units that moves with the market inside bounded ranges.

MU faced a similar type of engineering problem and was briefly cut from Rubin flagship configuration entirely and relegated to lower-tier Rubin chips. This illustrates how MU's position is more binary than SK's ("qualify on time or get cut"). In totality, this provides SK with much stickier relationships with its biggest customers—particularly with NVDA, which accounts for 27 percent of SK's revenue—because while competing suppliers face the same qualification risks, SK's four-generation incumbency gives NVDA strong incentive to wait out any stumbles rather than spend more to retool around a less-proven alternative.

The Rubin cycle also exposed deeper operational advantages at SK beyond incumbency. The base chip at the bottom of the HBM stack, which controls data flow between the GPU and the memory layers above, is a critical piece of logic best built on a leading-edge process node. SK and Samsung outsourced its fabrication to TSMC, while MU built it in-house on a process optimized for memory rather than logic. When NVDA raised HBM4 speed requirements midcycle, SK and Samsung could comply, but MU could not at viable yields and was cut from the Rubin flagship configuration and widely criticized for the cost-cutting decision that caused the miss.

Valuation Analysis

Similar to MU, SK's trailing and forward P/E multiples reflect a significant disparity given massive share price gains that have partly preceded earnings growth. Currently, shares of SK trade for 15 times trailing EPS, compared to a 2-year average valuation of 14x, and for 4.5 times NTM EPS against a 2-year average of 5.8x. Consensus projects earnings growth in excess of 55 percent for the coming year, driven by continued HBM volume ramp, sustained DRAM pricing, and operating leverage on a fixed cost base, making the current NTM P/E of

4.5x one of the more attractive entry points in the global semiconductor sector on a forward basis.

What makes the SK setup particularly compelling—and persistently misunderstood—is the scale of its discount relative to MU. SK holds a dominant 58 percent HBM market share versus MU's 21 percent, has a multi-generation incumbency with NVDA, and generates superior operating margins on its HBM business. Yet it trades at roughly 14x trailing earnings against MU's ~19x. That gap likely reflects the "Korea discount," a well-documented structural phenomenon in which Korean equities trade at a persistent valuation haircut to global peers. Additionally, SK trades only on the Korean KOSPI index with no US-listed ADR, which excludes a large share of global institutional capital that is either prohibited from or operationally unable to trade Korean-listed securities directly. The result is a company that the market treats more like a Korean industrial conglomerate than a dominant global AI infrastructure supplier.

HEADER REMOVED

SENSITIVITY ANALYSIS REMOVED

That said, the Korea discount is not purely irrational, and investors should understand what they are accepting. On the governance side, the chaebol structure gives SK's controlling family effective veto power over capital allocation, and the group has a record of related-party transactions and family-interest spending that institutional investors price as expropriation risk. Beyond these structural risks, SK faces the same memory downcycle exposure as MU, with the added wrinkle that its greater earnings leverage to HBM means any pricing collapse hits the income statement harder. In a risk-off environment, the governance and geopolitical haircuts tend to widen rather than narrow, so SK would likely underperform MU on a pure price basis even if the fundamental damage were symmetric. Investors with high conviction on the memory cycle and tolerance for the structural discount should find SK the more attractive risk-adjusted entry, while those uncertain on the cycle should be aware that SK's upside and downside are both amplified relative to the US-listed alternative.

HEADER REMOVED

Exhibit 1: Monthly Memory Pricing Trends

Monthly Memory Contract Pricing Trends										
Memory Type	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26
DDR4 1Gx8 Contract	\$2.00	\$2.00	\$2.09	\$3.17	\$3.17	\$3.17	\$4.50	\$7.50	\$8.00	\$13.00
Month/Month Change (%)	3%	0%	4%	52%	0%	0%	42%	67%	7%	63%
DDR5 1Gx8 Contract	\$2.50	\$2.50	\$2.50	\$2.63	\$2.68	\$2.68	\$3.23	\$3.45	\$3.45	\$7.56
Month/Month Change (%)	2%	0%	0%	5%	2%	0%	21%	7%	0%	119%
NAND TLC Flash 512Gb Contract	\$2.90	\$3.00	\$3.00	\$3.12	\$3.12	\$3.12	\$4.50	\$5.86	\$9.60	\$16.00
Month/Month Change (%)	12%	3%	0%	4%	0%	0%	44%	30%	64%	67%

Source: InSpectrum Tech Inc, Bloomberg Intelligence

Bloomberg Intelligence 

Exhibit 2: Capex Projections for the Big 4 Hyperscalers

